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Merrivoss Bioanalysis Inc

Matrix acceptance sign-off

Bioanalytical QA, signed copy

Merrivoss Bioanalysis Inc, Madison

Prepared by: Helen Marchwood, Bioanalytical Quality Manager

Date: 09/13/2011

1 Purpose

This sign-off records review responsibilities for tandem-MS matrix acceptance. File it with the current assay worksheet template and apply the acceptance criteria in the analyte-specific method.

2 Calibration checks

Each run carries a calibration curve prepared in the matrix of the study samples. The regression model and weighting are those specified in the analyte-specific method. Back-calculated calibrator concentrations, permitted exclusions and the remaining curve must meet that method's acceptance criteria before sample concentrations are released.

Table 1 Calibration and matrix checks applied to each run

Check	Applied to	Action if it fails
Curve fit and back-calculation	Every calibration point	Investigate the failure; any permitted exclusion and refit is documented, and the remaining curve must meet method acceptance criteria
Blank matrix and carry-over checks	Matrix blanks and sequence blanks at method-defined positions	Hold affected results while investigating interference or carry-over; apply the method acceptance criteria
Internal standard response	Every sample	Investigate the response and sample preparation; reanalyse where justified and document the decision
Quality control placement	Start, middle and end of each batch	Batch rejected and repeated if the run-level QC fails

Precursor-to-product ion transitions are confirmed against the reference standard before a new analyte card is opened. Where a qualifier transition is part of the method, a response ratio outside its tolerance triggers investigation of identity, interference and integration. Affected results remain on hold until the discrepancy is resolved and documented; the quantitation transition alone does not override the identity check.

3 Batch review responsibilities

The analyst prepares the batch and completes the first review. A second scientist, not involved in the run, checks the calibration, QC and internal standard data before the batch is released to the study director.

- Analyst: batch preparation, first review, worksheet entries
- Second reviewer: independent check of calibration and QC, signed on the worksheet
- Mass-spectrometry manager: instrument status and any tuning changes
- Study director: release of data to the study report, after matrix acceptance is recorded

Reviews are signed and dated on the worksheet in ink. Electronic reviews are accepted only where the LIMS audit trail is available for the same run.

4 Control lot reference

The assay worksheet must identify the control preparation and its source lot references, matching the preparation record and storage log. Qualification of a replacement control preparation follows the analyte-specific method; the reviewer checks that record before the preparation is used for run acceptance.

Correction to the control lot reference in the draft of this sign-off has been made by hand on the signed copy and is initialled below.

5 Sign-off

Role	Name	Date
Bioanalytical Quality Manager	Helen Marchwood	09/13/2011
DMPK Lead	Matteo Rinalvesi	09/13/2011
Mass-Spectrometry Manager	Javier Quintana	09/13/2011